

Robust poling and frequency conversion on thin-film periodically poled lithium tantalate

Cite as: APL Photon. 10, 091302 (2025); doi: 10.1063/5.0272629

Submitted: 14 March 2025 • Accepted: 30 August 2025 •

Published Online: 15 September 2025



Anna Shelton,^{a)} C. J. Xin, Keith Powell, Jiayu Yang, Shengyuan Lu, Neil Sinclair, and Marko Loncar

AFFILIATIONS

John A. Paulson School of Engineering and Applied Sciences, Harvard University, 9 Oxford Street, Cambridge, Massachusetts 02138, USA

^{a)} Author to whom correspondence should be addressed: annashelton@g.harvard.edu

ABSTRACT

We explore a robust poling method for periodically poled thin-film lithium tantalate (PP-TFLT) by systematically varying poling parameters and confirming the quality of inverted domains with second-harmonic microscopy. We find a periodic poling process that is insensitive to acoustic-grade or optical-grade thin film, electrode material, and the presence of an oxide interlayer. By using a single high-voltage electrical pulse with a peak voltage time of 100 ms or less and a ramp-down time of 90 s, rectangular poling domains are established and stabilized in the PP-TFLT. We employ our robust periodic poling process in a pole-after-etch approach to produce PP-TFLT ridge waveguides with normalized second-harmonic generation (SHG) conversion efficiencies of $208\% \text{ W}^{-1} \text{ cm}^{-2}$ from 1550 to 775 nm, which is in line with the theoretical value of $244\% \text{ W}^{-1} \text{ cm}^{-2}$. This work establishes a reliable poling process and demonstrates SHG from the telecommunications band to the near-visible band using thin-film lithium tantalate, expanding the capabilities of the platform for frequency mixing applications in quantum photonics, sensing, and spectroscopy.

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I. INTRODUCTION

Thin-film lithium tantalate (TFLT) has garnered increasing attention in recent years as an attractive alternative to the more mature platform of thin-film lithium niobate (TFLN) for applications in photonic integrated circuits.^{1–4} Similarly to TFLN, TFLT is a ferroelectric crystal offering a wide transparency window, a large electro-optic response ($r_{33} \approx 30 \text{ pm/V}$), and a large second-order nonlinearity ($d_{33} \approx -21 \text{ pm/V}$). Moreover, TFLT also features dramatically reduced birefringence,² enhanced DC stability,^{5,6} and, in bulk, has demonstrated a larger bandgap and a superior threshold for the photorefractive effect.⁷ Collectively, these properties have enabled the rapid development of nearly octave-spanning electro-optic frequency combs,² Kerr combs,¹ and electro-optic modulators that support stable DC biasing on the timescale of days.⁶ In contrast, periodically poled TFLT (PP-TFLT) is only starting to emerge.^{8–10}

Periodic poling is the periodic reversal of the domain orientation achieved in TFLN and TFLT through the application of a strong, localized electric field. A number of devices have been

demonstrated on periodically poled TFLN (PP-TFLN),^{11–13} enabling frequency up and down conversion of interest in, for example, quantum photonics,^{14–17} sensing,¹⁸ and spectroscopy.¹⁹ While its normalized conversion efficiency may be lower due to its lower second-order nonlinearity than TFLN, PP-TFLT may offer important advantages in near-visible and shorter wavelength applications due to its broader transparency window and lessened photorefractive effects.^{20,21} However, the periodic poling process for PP-TFLT has remained a challenge. Prior work on PP-TFLT has relied on poling pulse schemes requiring the simultaneous optimization of electric field strength, pulse number, and pulse length across a broad parameter space, where poling is typically performed prior to waveguide fabrication.^{8,10} Here we explore a robust poling process for PP-TFLT by systematically varying poling parameters and confirming the quality of inverted domains with second-harmonic microscopy (SHM). We find a periodic poling method that achieves ideal poling with the optimization of only one parameter, poling voltage. In addition, we find that the poling method is insensitive to the grade of TFLT used (acoustic-grade or optical-grade), poling

electrode material, and the presence of an oxide interlayer. These parameters are known to impact the poling quality of PP-TFLN, on the other hand.^{22–25} We then employ the poling pulse in a pole-after-etch process²⁶ to produce PP-TFLT waveguides with normalized conversion efficiencies of $208\% \text{ W}^{-1} \text{ cm}^{-2}$ from telecommunications to near-visible wavelengths, which is in line with the theoretical value of $244\% \text{ W}^{-1} \text{ cm}^{-2}$.

II. ROBUST PP-TFLT PROCESS

We begin our investigation of poling on PP-TFLT with the conventional poling process, pole-before-etch, where poling is performed on the unetched TFLT film before a waveguide for frequency conversion may be fabricated. The pole-before-etch process allows for fast turnaround times between samples since it does not require fabrication of the optical waveguide. Instead, it requires only one lithography step, followed by metal deposition and lift-off to define poling electrodes, and thus is ideal for rapid evaluation of the

optimal poling pulse parameters. Our device is fabricated on a $1 \times 1.5 \text{ cm}^2$ TFLT chip featuring a 500-nm-thick *x*-cut LT device layer on top of 2000-nm-thick SiO_2 on a Si handle wafer. The fabrication starts by cleaving the chip out of an optical-grade TFLT wafer (NanoLN), followed by a solvent clean. Electrodes are patterned using Micro-Chem LOR and Zeon Chemicals ZEP520 resists. Rounded-tip electrode fingers are $2 \mu\text{m}$ in width and $25 \mu\text{m}$ in length for a $6 \mu\text{m}$ poling period. Electrode pads, used to land electric probes during the poling process, are $\sim 100 \mu\text{m}$ in width and are electrically connected to each poling finger. The high-voltage and ground electrode fingers are separated by a gap of $15 \mu\text{m}$. After the lithography step, an oxygen descum is performed, followed by the evaporation of 50 nm of chrome and 100 nm of gold as a cap layer. Lift-off is performed, and photoresist is spun on top of the chip to prevent the electrical breakdown of air during the poling process.

Our poling method is a single pulse with the profile shown in Fig. 1(b). The pulse consists of a 1 kV/ms linear ramp-up that increases the voltage from 0 to the peak voltage, and the peak voltage

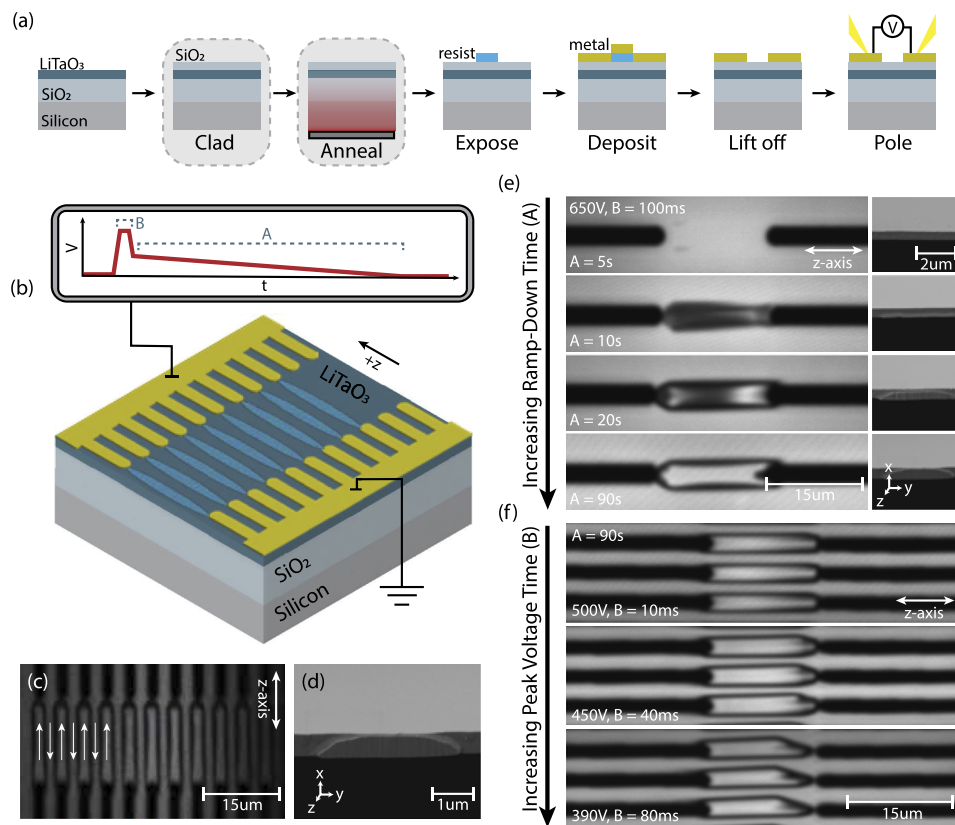


FIG. 1. (a) Fabrication flow for the robust PPLT study, with optional steps shown in gray dashed boxes. (b) Illustration of a PPLT sample during the poling process with a qualitative poling pulse form highlighting relevant pulse parameters. A is the ramp-down time. B is the peak-voltage time. (c) SHM image of a PPLT sample with a near 50% duty cycle. Black rectangles at the top and bottom of the image are poling fingers. Dark lines between the fingers indicate domain walls. Bright regions indicate where the crystal domains of the TFLT are aligned. Arrows indicate the direction of the crystal domain for each bright region. (d) SEM cross-sectional image of a PPLT sample after the differential etch to expose domain orientation. The domain reaches 100% depth in the LT film. (e) SHM images (left) and SEM cross-sectional differential etch images (right) of PPLT samples with varying pulse ramp-down times. Pulse details are noted on each SHM image. As pulse ramp-down time (A) increases, poled domains penetrate deeper into the LT film. (f) SHM images of PPLT samples optimized for varying peak voltage hold times. Pulse details are noted on each SHM image. As peak-voltage time (B) increases, domains become more asymmetric.

is held for 100 ms. The peak voltage is allowed to vary to determine the duty cycle of the poled region. Then, a linear 90-s-long voltage ramp-down is used, starting at half of the peak voltage [Fig. 1(b)]. Poled domains are then inspected with SHM to assess duty cycle and poling depth [Fig. 1(c)]. The SHM is a home-built system consisting of a Newport InSight DeepSee femtosecond laser (wavelength: 795 nm, pulse width: 110 fs, and average power: 375 mW) used with a Leica SP5 confocal microscope. The reflected signal is detected by a Becker and Hickl time-correlated single-photon counting GaAsP detector. Bright regions in the SHM indicate regions where the crystal domains of the TFLT are aligned; dark regions indicate boundaries between regions of opposite crystal domain orientation; and electrodes appear as solid black shapes in the SHM. When regions between poling fingers are bright and bounded by dark boundaries, domain inversion has taken place. When the width of the domain inversion region reaches 50% of the poling period, a 50% duty cycle is obtained, which is ideal for achieving maximum nonlinear frequency conversion.²⁷ To confirm SHM images, windows are opened via optical lithography on both sides of the center line of each poled region perpendicular to the poling fingers. We then perform an argon-based ICP-RIE etch through the full thickness of the TFLT and chemical redeposition removal process using heated KOH and SC-1. Finally, a 20-minute 49% HF wet etch is performed, which preferentially etches the $-z$ face of lithium tantalate,²⁸ showing poled regions reach 100% depth in the TFLT [Fig. 1(d)].

By adjusting the peak voltage, a single poling pulse with the profile shown in Fig. 1(b) successfully poles our device and reaches a nearly 50% duty cycle at 650 V. Holding all other pulse parameters stable, for peak voltages above 650 V, domains broaden and eventually merge; for peak voltages below 650 V, domains narrow, become more shallow, and eventually no nucleation can be observed. Other phenomena were also observed. Shortening the ramp-down time resulted in increasingly shallow domains [Fig. 1(e)], with ramp-down times less than 10 s resulting in near-uniform backswitching of the TFLT between poling fingers. This pulse shape was inspired by studies on bulk PPLT,²⁹ where increased time is necessary for domain stabilization to avoid domain backswitching. Poling depth was improved by increasing ramp-down time up to 90 s. Beyond 90 s, no further enhancement of poling depth was observed. In addition, some devices formed triangularly shaped domains. By shortening the pulse peak-voltage time from 100 to 10 ms or shorter and reoptimizing only the poling field strength, rectangular domains were recovered [Fig. 1(f)].

Having shown our poling method works on one device, we sought to assess how robust the method is to different sample preparations. Fabrication conditions such as film grade, contact metal, and oxide interlayer are known to influence poling on TFLN. Considering film grade, film grade dictates the coercive field of the sample in bulk PPLT, with samples closer to stoichiometry typically having lower coercive fields.³⁰ For electrode contacts, contact metal is related to nucleation density in bulk PPLN.²² Perhaps most interesting for its applications to UV up-conversion, an oxide interlayer reduces leakage current through the charged domain walls, improving lateral domain spreading and merging for small poling periods in PP-TFLN.²³ In addition, annealing may change the poling conditions. Annealing TFLN at either 300 or 520 °C has been shown to mitigate electron-beam-induced defects,²⁴ but it also changes the DC

TABLE I. Summary of PP-TFLT devices fabricated in our study as well as the gap between poling electrodes and the electric field applied to pole each device. In all cases, by adjusting only one parameter—the peak voltage—a poling pulse with a 90 s ramp-down time and a 100 ms peak-voltage time successfully poles each device. Our material stack consists of a 500-nm-thick x-cut TFLT device layer on top of 2000-nm-thick SiO₂ on a Si handle wafer (NanoLN). Devices consist of different TFLT film grades (acoustic or optical), oxide interlayers, annealing treatments, poling electrode contact metals, and poling finger gaps. Importantly, we find the optimal poling recipe is mostly insensitive to these variations, and all samples pole when a poling field strength of ~50 kV/mm is used. We note that higher poling field strengths are required when an oxide interlayer is added and annealed in an ambient environment at 300 °C for 8 h.

Grade	SiO ₂ (nm)	Annealing	Contact	Gap (μm)	kV/mm
Optical	...	...	NiCr	10	41.5
Acoustic	...	...	Ni	15	42
Acoustic	...	...	Cr	15	43
Optical	80	520 °C, 5 h	Ni	15	43
Optical	...	...	Cr	15	43
Optical	...	...	Ni	15	43
Optical	...	...	Ti	10	50
Optical	...	...	Ni	10	50
Optical	50	300 °C, 8 h	Ni	15	53
Optical	100	300 °C, 8 h	Ni	10	70

electrical conductivity.²⁵ To test the efficacy of our poling method against these conditions in TFLT, we fabricated samples with similar variations and attempted to pole.

The fabrication flow was the same as that of the previous devices with a few notable modifications: both acoustic-grade and optical-grade TFLT wafers were used, and on selected chips, an oxide interlayer was deposited via plasma-enhanced chemical vapor deposition (PECVD) and annealed either in an oxygen atmosphere at 520 °C for 5 h or in ambient air at 300 °C for 8 h [Fig. 1(a)]. The poling electrodes consisted of 50–100 nm of a test metal (chrome, nickel, titanium, or nichrome), followed by a 100 nm gold capping layer.

We then apply our poling method, adjusting only the peak voltage to adjust the duty cycle. We find that, for each combination of fabrication conditions, periodic poling with a nearly 50% duty cycle was achieved (Table I). In addition, as the electrode gap increases, the coercive field tends to decrease, following the Janovec–Kay–Dunn (JKD) scaling law. Because we were able to pole all samples successfully using the same pulse shape and poling procedure, our method is robust to a wide range of sample conditions. Crucially, the only parameter that needs to be adjusted is the poling voltage, which accounts for variations such as the presence of an insulating oxide interlayer or local differences in film thickness and crystal quality. This one-parameter optimization enables reliable periodic poling across PP-TFLT platforms without further tuning of the pulse waveform.

III. SECOND HARMONIC GENERATION

We utilize a pole-after-etching approach²⁶ to fabricate an efficient PP-TFLT ridge waveguide for second-harmonic generation (SHG) from 1550 to 775 nm. Importantly, we use the same poling

pulse shape as the one discussed in the previous section and only adjust the peak voltage we apply while keeping all other parameters the same. A pole-after-etch approach offers two key advantages: increased yield of devices operating at a desired target wavelength and foundry compatibility. In this approach, the waveguides are first etched, and the fabricated sample is characterized using ellipsometry to measure etch depth variations. Then, metal electrodes used for poling are deposited, with periodicity adjusted to account for the film thickness and etch depth variations. Due to the sensitivity of the phase-matching function to geometric dispersion, etching first allows the waveguide geometry to be measured and fabrication-induced variations to be taken into account. This significantly increases the probability of fabricated frequency converters operating at desired wavelengths, thus increasing the yield of the process. Furthermore, since poling is a specialized technique and not a standard process available in commercial foundries used to fabricate TFLN and TFLT devices, moving it to the back end of line (BEOL) is desirable. In this approach, poling is done after etching (after all wafer-scale processing is completed), thus making PP-TFLT compatible with large-scale manufacturing. One drawback of a pole-after-etch approach is lower nonlinear conversion efficiency when compared to a pole-before-etch approach due to the lack of domain inversion in the ridge, resulting in poling of only the bottom portion (slab) of the waveguide. However, this can be addressed by the recently demonstrated sidewall poling approach.³¹ When a pole-after-etch approach is used, the electric field in the waveguide ridge is insufficient for nucleation, and domain wall speed along the x -axis is too slow for permeation, so poled domains do not start or penetrate

deeply into the waveguide ridge.³² For our work, we seek a reliable poling method capable of being employed in BEOL fabrication, so we opt for the pole-after-etch method in realizing SHG.

We begin fabrication by cleaving chips of size $1.8 \times 1.8 \text{ cm}^2$ from a 500-nm-thick x -cut LT device layer on top of 2000-nm-thick SiO_2 on a Si handle wafer and perform a solvent clean [Fig. 2(a)]. We perform ellipsometry on the initial film to obtain the average thickness of the device layer, then use high-resolution electron beam lithography with ma-N 2400 resist to define $2.5 \mu\text{m}$ wide (top-width) and 1.4 cm long waveguides. The chip then undergoes an argon-based ICP-RIE etch to etch $\sim 300 \text{ nm}$ of the TLFT and a chemical redeposition removal process using hot KOH and SC-1. Finally, ellipsometry is performed again to evaluate the average thickness of the remaining (unetched) film. The etch process produces a sidewall angle of 61° [Fig. 2(d)]. By fabricating the ridge waveguide before poling, actual device parameters can be used to calculate an accurate poling period and thus overcome thickness variations and fabrication imperfections. The measured film thickness, etch depth, and sidewall angle are input into simulation to extract the wavevector for the fundamental transverse electric mode at 1550 nm (k_{th}) and 775 nm (k_{sh}). The poling period (Λ) is then determined by

$$\Lambda = \frac{2\pi}{k_{\text{sh}} - 2k_{\text{th}}} = 3.908 \mu\text{m}. \quad (1)$$

Poling electrodes are fabricated identically to the first test device, but the poling period is set to the simulated value of $3.908 \mu\text{m}$, and the rounded poling finger width is set to $1.303 \mu\text{m}$.

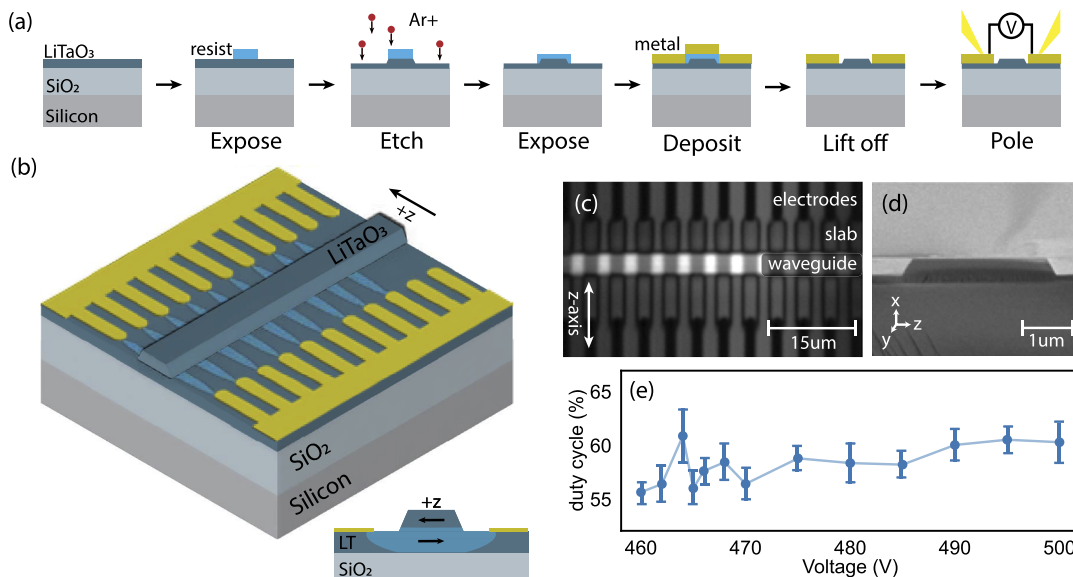


FIG. 2. (a) Fabrication flow for the PP-TFLT chip for SHG. (b) Illustration of a PPLT sample with electrodes deposited and poling performed after the ridge waveguide etching process. Cross-sectional illustration (bottom right) of a poled domain for a PPLT sample prepared as shown. The poled domain does not extend far upward into the ridge waveguide. (c) SHM image of this PPLT sample with a near 50% duty cycle. Black rectangles at the top and bottom of the image are poling fingers. For poling on the slab, dark lines between the fingers indicate domain walls. Bright regions indicate where the crystal domains of the TFLT are aligned. Poling on the waveguide ridge appears differently due to its lessened depth. For poling on the waveguide ridge, dark regions indicate poled regions, and bright regions indicate unpoled regions. The poled domains align across the slab and waveguide between each poling finger. (d) SEM of a PPLT waveguide facet showing a clean facet and 61° sidewall angle. (e) Duty cycle as a function of peak voltage for this PPLT sample with a 5 ms peak-voltage hold time and a 90 s ramp-down time.

The electrode region is set to a total length of 1 cm to improve output second-harmonic power [Fig. 2(b)]. To optimize the pole-after-etch process, we start with the same pulse shape found to be effective in the pole-before-etch process and sweep a wide range of peak voltage values. Due to the presence of triangular domains, we also use a shorter peak-voltage time of 5 ms and optimize the duty cycle of our device by adjusting the peak voltage [Fig. 2(e)]. At 460 V, rectangular domains with a duty cycle of 55% are found by SHM imaging [Fig. 2(c)]. We find the peak voltage needed for the pole-after-etch device (460 V) is lower than that used in pole-before-etch devices with the same electrode gap (645 V). After poling, poling electrodes are removed by gold etchant and chrome etchant. Chips are then singulated by a fluorine- and argon-based ICP-RIE etch and Bosch etch before measurement.³³

The PP-TFLT chip is measured using an end-fire coupling setup with pump light from a continuous-wave telecommunication-wavelength tunable laser (Santec TSL-710) in the range of 1480–1640 nm [Fig. 3(b)]. Pump light passes through a polarization controller and couples into the chip by lensed fiber. The resultant near-visible second-harmonic light is collected at the output by a lensed fiber and is measured using a visible-wavelength photodetector. As the laser wavelength is swept into the designed quasi-phase matched wavelength, the second-harmonic output power rapidly maximizes. The experimental conversion efficiency (η_{exp}) can be calculated as

$$\eta_{\text{exp}} = \frac{P_{\text{out}}}{P_{\text{in}}^2 L^2}, \quad (2)$$

where P_{out} is the output second-harmonic power, P_{in} is the fundamental on-chip power, and L is the length of the poled region [Fig. 3(a)]. The calibration method measures the output optical power at the second-harmonic wavelength, output optical power at the fundamental-harmonic wavelength, and internal power read-out of the laser. Setup losses are removed, and coupling losses are

calculated by assuming the input and output losses are equal, as both input and output facets are visually identical, and propagation losses are negligible because the waveguides are 1.2 cm in length and similarly fabricated devices have losses of 9 dB/m⁶. On-chip fundamental and second-harmonic powers are determined from the recorded input and output data and the calculated coupling losses. These corrected values are then used to compute the normalized SHG efficiency. After setup and facet loss calibration is performed, the maximum experimental conversion efficiency achieved on the PP-TFLT chip is 208% W⁻¹ cm⁻² at 1552 nm.

Because we use a pole-after-etch approach, poling electrodes are deposited onto the waveguide slab, and the electric field induced by the poling electrodes is insufficient to reverse the polarization direction in the uppermost portion of the ridge waveguide. To ascertain poled domain depth, windows were opened via optical lithography to bisect the ridge waveguide parallel to the central line. We then perform the same argon-based ICP-RIE etch through the full thickness of the TFLT, chemical redeposition removal process using heated KOH and SC-1, and 20-minute 49% HF wet etch as done in our robust poling study, revealing domains only extend through the bottom 178.7 nm of TFLT [Fig. 3(c)]. The peak theoretical conversion efficiency (η_{theory}) can be calculated for this waveguide geometry and experimentally measured depth as

$$\eta_{\text{theory}} = \frac{2\omega^2 d_{\text{eff}}^2}{n_{\text{fh}}^2 n_{\text{sh}} \epsilon_0 c^3} \frac{A_{\text{sh}}}{A_{\text{fh}}^2}, \quad (3)$$

where ω is the fundamental frequency, d_{eff} is given by the second order nonlinear tensor, n_{fh} is the refractive index of the fundamental TE mode at the fundamental frequency, n_{sh} is the refractive index of the fundamental TE mode at the second harmonic, A_{fh} is the fundamental TE mode area at the fundamental frequency, A_{sh} is the fundamental TE mode area at the second harmonic, ϵ_0 is the permittivity of free space, and c is the speed of light, resulting in a peak theoretical conversion efficiency of 244% W⁻¹ cm⁻². This

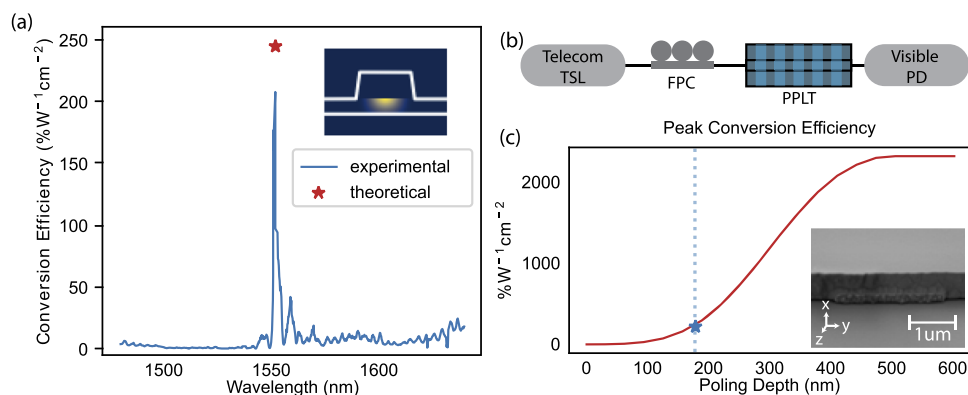


FIG. 3. (a) Normalized conversion efficiency (line) as a function of wavelength for a PPLT device, showing a maximum value of 208% W⁻¹ cm⁻² at 1552 nm. The peak theoretical normalized conversion efficiency for this device is 244% W⁻¹ cm⁻² at 1552 nm (star). The inset shows the effective optical mode area for QPM. Since the poling only penetrates the slab of the waveguide, only the optical mode in the slab experiences the periodic poling for phase matching. (b) Illustration of the experimental setup for measuring SHG from PPLT. (c) Peak theoretical normalized conversion efficiency as a function of poled domain depth in the LT film for an identical geometry to the measured device. The poling depth of this device is highlighted by a vertical line, and the experimental peak normalized conversion efficiency is plotted (star). The inset shows an SEM cross-sectional image of this PPLT sample after the differential etch to expose domain orientation. The domain reaches 178.7 nm depth in the LT film.

value agrees well with our experimental findings. Poling depth can be improved by sidewall poling.³¹ With poling depth equal to the full film thickness for the same waveguide geometry as our experimental ridge waveguides, like in the case of the pole-before-etch method, peak conversion efficiency can reach 2314% $W^{-1} cm^{-2}$ [Fig. 3(c)]. The most efficient demonstration of frequency conversion in PP-TFLT has achieved 1290% $W^{-1} cm^{-2}$ peak conversion efficiency⁹ and in PP-TFLN has surpassed 4600% $W^{-1} cm^{-2}$ peak conversion efficiency.³⁴ Had our device been poled to 100% depth, it may exceed the performance of the state-of-the-art device for frequency conversion in PP-TFLT, but due to its lower second-order nonlinearity, even the most efficient PP-TFLT has yet to achieve conversion efficiencies as high as PP-TFLN.

IV. CONCLUSION

We found a periodic poling method that achieves high-quality poling with the optimization of only one parameter—poling voltage—and is agnostic to acoustic-grade and optical-grade TFLT, oxide interlayers, and poling electrode metals. We also experimentally confirm that the poling method can be performed in both a conventional pole-before-etch and a pole-after-etch approach. For each sample preparation tested, periodic poling with a duty cycle near 50% was achieved through the optimization of the poling voltage alone, demonstrating how robust our poling process is. We also realized a PP-TFLT ridge waveguide using a pole-after-etching process, featuring a normalized conversion efficiency of 208% $W^{-1} cm^{-2}$. This value matches closely with the theoretical normalized conversion efficiency of 244% $W^{-1} cm^{-2}$. By increasing the depth of periodically poled regions in TFLT, the theoretical conversion efficiency can approach 2314% $W^{-1} cm^{-2}$ for an identical waveguide geometry. Due to its higher threshold for photorefractive DC stability, and wider bandgap, this work paves the way for PP-TFLT to address high-power and high-frequency applications in quantum optics, sensing, and spectroscopy.

ACKNOWLEDGMENTS

The authors thank Cornelis (Kees) Franken and Soumya Ghosh for discussions related to the project. The authors thank Meg Reardon and Ganna Savostyanova for administrative support. K.P. acknowledges support from the Harvard GRID Fellowship. S.L. acknowledges support from the A*STAR National Science Scholarship. This work was supported by Amazon Web Services (Grant No. A50791), the Air Force Office of Scientific Research (Grant No. FA9550-23-1-0333), the National Research Foundation of Korea (Grant No. NRF-2022M3K4A1094782), NASA (Grant Nos. 80NSSC22K0262 and 80NSSC23PB442), the National Science Foundation (Grant Nos. OMA-2137723, EEC-1941583, and 213806), and the Office of Naval Research (Grant No. N00014-22-C-1041). This work was performed in part at the Harvard University Center for Nanoscale Systems (CNS), a member of the National Nanotechnology Coordinated Infrastructure Network (NNCI), which is supported by the National Science Foundation under NSF Award No. ECCS-2025158.

AUTHOR DECLARATIONS

Conflict of Interest

K.P., N.S., and M.L. are involved in developing lithium tantalate technologies at Lumina Corporation. M.L. is involved

in developing lithium niobate technologies at HyperLight Corporation.

Author Contributions

A.S. designed the devices, performed the experiments, and analyzed the data. A.S., C.J.X., and K.P. developed the fabrication process. A.S. and K.P. fabricated the devices. J.Y. and S.L. provided advice on the fabrication process and experiment. A.S., M.L., and N.S. wrote the manuscript with comments from all authors. M.L. supervised the project.

Anna Shelton: Data curation (lead); Formal analysis (lead); Investigation (lead); Methodology (equal); Writing – original draft (lead). **C. J. Xin:** Formal analysis (supporting); Methodology (equal). **Keith Powell:** Conceptualization (lead); Methodology (equal). **Jiayu Yang:** Methodology (supporting). **Shengyuan Lu:** Methodology (supporting). **Neil Sinclair:** Writing – review & editing (equal). **Marko Loncar:** Funding acquisition (lead); Supervision (lead); Writing – review & editing (equal).

DATA AVAILABILITY

The data that support the findings of this study are available from the corresponding author upon reasonable request.

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